Rank-27 Search-Quality Tie Closure

Tactix Development Project

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Outcome

The first later decision mismatch after the previously closed equal-score choice is itself another equal-score heap-topology tie. Exact candidate admission, activity values, score-bump work, phases, and learning provide no worse-side Tactix owner. The arm is closed without a solver change.

1 Exact observer

Array `1650364` ran two independent solver tasks simultaneously on node018. Each task owned one solver invocation, 12 allocated logical CPUs, exactly 48,000 MiB scheduler memory, and a 16 GiB process cap. Both tasks completed `0:0` in 33 seconds. Tactix exited zero at its terminal hook; the current-Kissat observer raised its accepted terminal SIGTERM and the wrapper classified expected exit 143 as success.

The current-source Tactix trace pin has SHA-256 `2b3a0d062268e1f7d7f1f19cc7336d915f8494fb1640afbfc8d10170474f2799`. The retained current-Kissat observer pin has SHA-256 `6c30f2d0d476e34ad471e951410881625216afe366db7cff26361012c4dab385`. Seven focused scheduling, strategy, trace, proof-boundary, and reset tests pass.

2 Decision classification

Both observers learn the same 16-literal clause at conflict 9459. They then emit exactly 87 score bumps in the same order; the ordered payload SHA-256 is `52031fda941a8be8a1050b9bb3a74876f7106360990183e1f3dbe498ea2cbda3` on both sides. Recorded decision 5,795,464 is the common literal `-117212`.

At recorded decision 5,795,465, Tactix chooses `-117204` and Kissat chooses `-117206`. Both candidate variables are admitted on both sides, and both carry the bit-identical activity value `364280533501.80402`. Both phases are negative, and the random-decision state is inactive. Only the pre-existing equal-score heap topology determines which candidate reaches the root first.

The two solvers subsequently traverse different members of this equal-score cohort and encounter different conflicts at conflict 9460. That is downstream of the neutral topology choices; it is not evidence of weaker scoring, candidate admission, phase selection, learning, pruning, or correctness. Changing the tie rule would be strategy tuning without a demonstrated worse-side owner and is therefore prohibited by the convergence contract.

3 Disposition

No solver behavior changes and no performance claim follows from this observer. Broad-search work returns to exact source-equivalent cost ownership outside all closed tie, propagation, collection, Vivify, Substitute, residual BVE, and Congruence-layout arms.

Both staged CNFs and node-local scratch directories are absent. The bounded UNKNOWN observers created no proof, model, checker, DRAT, LRAT, or certificate artifact.